



TIRZEPATIDE: PHARMACOLOGY, CLINICAL EVIDENCE, AND THERAPEUTIC POTENTIAL

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CERTIFICATE

This is to certify that Mr. **Sayan Ghosh** has successfully completed the Project Work (PT883) work entitled "**TIRZEPATIDE: PHARMACOLOGY, CLINICAL EVIDENCE, AND THERAPEUTIC POTENTIAL**" in partial fulfillment of the requirements for the award of the degree of Bachelor of Pharmacy (B.Pharm.) during the 8th Semester at BCDA College of Pharmacy & Technology, affiliated to Maulana Abul Kalam Azad University of Technology (MAKAUT), Kolkata.

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ABSTRACT

Background: Tirzepatide is a novel dual glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptor agonist developed by Eli Lilly and Company for the management of type 2 diabetes mellitus (T2DM) and obesity. Its unique dual incretin mechanism distinguishes it from existing GLP-1 receptor agonists such as semaglutide and liraglutide.^[1,2]

Objective: This review comprehensively evaluates the pharmacology, pharmacokinetics, clinical efficacy, safety profile, and therapeutic potential of tirzepatide based on available peer-reviewed evidence.^[3]

Methodology: A systematic literature search was conducted across PubMed, MEDLINE, NEJM, The Lancet, JAMA, and ClinicalTrials.gov, using the search terms 'tirzepatide', 'GIP/GLP-1 agonist', 'SURPASS', and 'SURMOUNT'. Publications from 2020–2026 were prioritised.

Major Findings: Across the SURPASS clinical programme (SURPASS-1 to SURPASS-5), tirzepatide (5, 10, 15 mg weekly) achieved HbA1c reductions of 1.87–2.59% and body weight reductions up to 11.4 kg versus placebo. The SURMOUNT programme demonstrated weight reductions of up to 22.5% in adults with obesity. Tirzepatide demonstrated superior glycaemic control and weight reduction compared to semaglutide 1 mg in the SURPASS-2 trial. Common adverse effects include nausea, diarrhoea, and vomiting, predominantly mild-to-moderate and transient.^[4,5,6]

Conclusion: Tirzepatide represents a significant pharmacological advancement in the management of T2DM and obesity, offering superior dual metabolic benefits over existing monotherapy approaches. Its once-weekly dosing schedule and cardiometabolic advantages reinforce its promising therapeutic role.^[7]

Keywords: *Tirzepatide; GIP receptor agonist; GLP-1 receptor agonist; Type 2 diabetes mellitus; Obesity; SURPASS; SURMOUNT; Dual incretin*

1. INTRODUCTION

The global burden of metabolic disease continues to escalate at an alarming pace. As of 2023, approximately 537 million adults worldwide were living with diabetes mellitus, a figure projected to reach 783 million by 2045.^[8] Obesity, a key driver of type 2 diabetes mellitus (T2DM), insulin resistance, and cardiovascular disease, affects over 1 billion individuals globally.^[9] Despite an expanding pharmacological armamentarium including metformin, sulphonylureas, dipeptidyl peptidase-4 (DPP-4) inhibitors, sodium-glucose cotransporter-2 (SGLT-2) inhibitors, and GLP-1 receptor agonists, sustained glycaemic control and durable weight reduction remain significant unmet clinical needs.^[10]

Existing GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated meaningful HbA1c reductions and moderate weight loss. However, limitations including incomplete weight reduction, gastrointestinal intolerance, and cost restrict their broader applicability.^[11] These therapeutic gaps have driven interest in agents targeting multiple incretin pathways simultaneously.

Tirzepatide (Mounjaro[®], Zepbound[®]) is a first-in-class once-weekly subcutaneous injection that acts as a dual agonist of both the glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptors.^[1] Developed by Eli Lilly and Company, it received United States Food and Drug Administration (FDA) approval for T2DM in May 2022 and for chronic weight management (obesity) in November 2023.^[12,13]

This review aims to provide a comprehensive, evidence-based appraisal of tirzepatide, covering its pharmacodynamics, pharmacokinetics, major clinical trial outcomes, safety profile, comparative analysis, and future therapeutic directions, to serve as a consolidated academic and clinical reference.^[3]

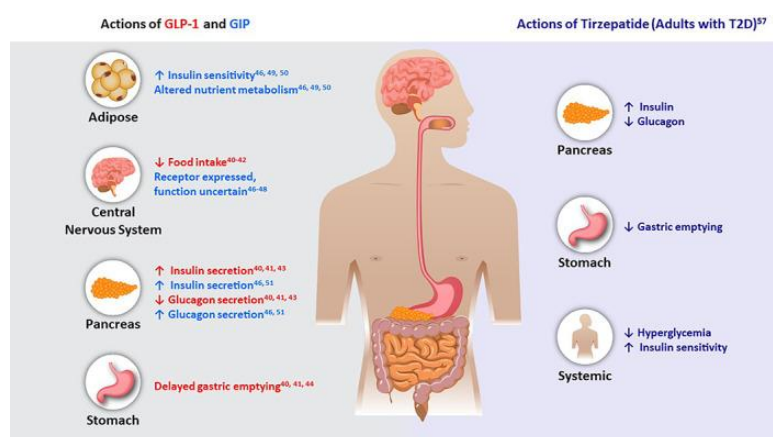


Figure1: Tirzepatide improves blood glucose control and promotes weight loss by activating both GIP and GLP-1 receptors in adults with type 2 diabetes.

2. DRUG OVERVIEW

Tirzepatide is a synthetic 39-amino acid peptide engineered to act simultaneously on both GIP and GLP-1 receptors. Its molecular backbone is derived from the native GIP sequence with strategic modifications to confer GLP-1 receptor binding affinity and prolonged pharmacokinetic stability.^[14]

Table 1: Drug Identity and Classification of Tirzepatide

Parameter	Details
Generic Name	Tirzepatide
Brand Names	Mounjaro® (T2DM); Zepbound® (Obesity/Weight Management)
Drug Class	Dual GIP/GLP-1 Receptor Agonist (Twincretin)
Chemical Classification	Synthetic 39-amino acid acylated peptide
Molecular Formula	C ₂₂₅ H ₃₄₈ N ₄₈ O ₆₈ S
Molecular Weight	~4813.5 Da
Developer / Manufacturer	Eli Lilly and Company, USA
Route of Administration	Subcutaneous injection (once weekly)
Available Doses	2.5 mg, 5 mg, 7.5 mg, 10 mg, 12.5 mg, 15 mg (autoinjector pen)
FDA Approval (T2DM)	May 13, 2022 (NDA 215866)
FDA Approval (Obesity)	November 8, 2023 (NDA 217806)
EMA Approval	September 2023 (Mounjaro®)
Storage	Refrigerated at 2–8°C; protect from light

The unique structural feature of tirzepatide is the C20 fatty diacid moiety attached via a gamma-glutamic acid linker, enabling high albumin binding and thus prolonging its half-life, supporting once-weekly subcutaneous administration.^[14,15]

3. PHARMACODYNAMICS AND MECHANISM OF ACTION

Tirzepatide exerts its pharmacological effects through simultaneous activation of both GIP and GLP-1 receptors, earning the designation 'twincretin'. This dual receptor engagement produces synergistic metabolic benefits exceeding those achievable by selective GLP-1 receptor agonism alone.^[1,2,16]

3.1 GLP-1 Receptor Activation

Tirzepatide binds to GLP-1 receptors expressed on pancreatic beta cells, hypothalamic neurons, and vagal afferent neurons. GLP-1 receptor activation stimulates glucose-dependent insulin secretion, suppresses glucagon release from alpha cells, delays gastric emptying, and promotes satiety via central appetite regulation.^[17,18] These mechanisms collectively reduce postprandial glucose excursions and caloric intake.

3.2 GIP Receptor Activation

Unlike GLP-1 receptor agonists, tirzepatide additionally activates GIP receptors. GIP receptor stimulation enhances insulin secretion in a glucose-dependent manner, improves insulin sensitivity in adipose tissue, and reduces adipogenesis.^[16,19] Critically, GIP receptor activation also attenuates the nausea and gastrointestinal adverse effects commonly associated with GLP-1 receptor agonism, potentially improving tolerability.^[20]

3.3 Cellular Signalling Pathways

Both GIP and GLP-1 receptors are G-protein coupled receptors (GPCRs) that primarily couple to Gs proteins. Upon agonist binding, adenylyl cyclase activation elevates intracellular cyclic AMP (cAMP), which:^[17]

- Activates protein kinase A (PKA), enhancing insulin exocytosis from beta-cell secretory granules^[17]
- Activates exchange protein activated by cAMP (EPAC2/Rap-GEF2), modulating calcium influx and insulin secretion^[17]
- Suppresses glucagon secretion from pancreatic alpha cells in a glucose-dependent manner^[18]
- Modulates appetite-regulating neuropeptides (PYY, NPY) in the hypothalamus^[21]

3.4 Weight Reduction Mechanisms

Tirzepatide achieves superior weight reduction through multiple complementary mechanisms:^[21,22]

- Hypothalamic GLP-1 receptor activation reduces appetite and promotes satiety signalling^[21]
- GIP receptor activation in adipose tissue reduces lipid accumulation and enhances lipolysis^[22]
- Delayed gastric emptying reduces postprandial appetite stimulation and energy intake^[18]
- Reduction in food cravings and caloric preference for high-fat foods^[22]

3.5 Metabolic Effects

Beyond glycaemic and weight effects, tirzepatide demonstrates broader cardiometabolic benefits including reductions in fasting serum triglycerides, LDL-cholesterol, total cholesterol, systolic blood pressure, and liver fat content.^[23] It also reduces markers of hepatic inflammation, raising interest in potential applications for non-alcoholic steatohepatitis (NASH).^[24]

3.6 Dose-Response Relationship

Tirzepatide exhibits a dose-dependent relationship for both HbA1c reduction and body weight loss. The 5 mg dose achieves clinically meaningful improvements, while the 10 mg and 15 mg doses confer progressively superior outcomes. Gastrointestinal adverse effects also show dose-dependent increases, necessitating a graduated dose-escalation protocol.^[4,5]

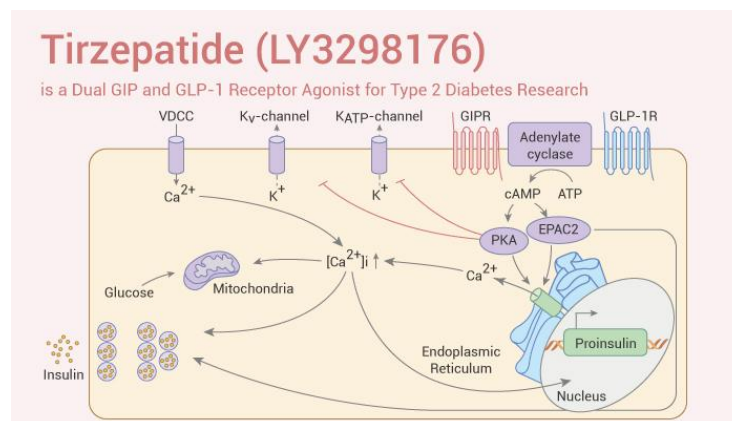


Figure 2: Mechanism of Action of Tirzepatide: Dual GIP and GLP-1 Receptor Agonism

Source: Adapted from Frias JP et al. [16] and Nauck MA et al. [17]

4. PHARMACOKINETIC PROFILE

The pharmacokinetic (PK) profile of tirzepatide has been characterised across phase I, II, and III clinical studies, demonstrating properties consistent with once-weekly subcutaneous dosing.^[25,26]

4.1 Absorption

Following subcutaneous injection, tirzepatide is absorbed with a median time to peak concentration (T_{max}) of approximately 8–72 hours (median ~8.5 hours). Absolute bioavailability after subcutaneous administration is approximately 80%, with no clinically meaningful difference across injection sites (abdomen, thigh, or upper arm).^[25]

4.2 Distribution

Tirzepatide is extensively bound to plasma albumin (>99%), which is the primary mechanism responsible for its prolonged half-life and supports weekly dosing.^[26] The volume of distribution at steady state is approximately 10.3 L, consistent with limited tissue distribution beyond the central compartment.

4.3 Metabolism

Tirzepatide undergoes proteolytic cleavage of the peptide backbone and beta-oxidation of the C20 fatty acid chain. It is not a substrate for cytochrome P450 (CYP450) enzymes, reducing the potential for pharmacokinetic drug interactions.^[25,26] Metabolites are pharmacologically inactive.

4.4 Elimination

The mean elimination half-life (t_{1/2}) of tirzepatide is approximately 5 days (116–134 hours), supporting once-weekly dosing. Steady-state plasma concentrations are achieved after approximately 4 weeks of weekly administration.^[25] Excretion occurs primarily via renal and faecal routes as degraded peptide fragments; no dose adjustment is required in patients with mild-to-moderate renal impairment.^[26]

Table 2: Summary of Pharmacokinetic Parameters of Tirzepatide

Parameter	Value	Clinical Significance
Bioavailability	~80%	Good subcutaneous absorption
Tmax	8–72 h (median ~8.5 h)	Gradual absorption profile
Protein Binding	>99% (albumin)	Supports long half-life
Volume of Distribution (Vd)	~10.3 L	Limited tissue distribution
Half-Life (t_{1/2})	~5 days (116–134 h)	Supports once-weekly dosing
Time to Steady State	~4 weeks	Dose escalation over 4-week intervals
Metabolism	Proteolysis; fatty acid beta-oxidation	No CYP450 interaction
Excretion	Renal and faecal (peptide fragments)	No dose adjustment in mild–moderate renal impairment
Clearance	~0.061 L/h	Low systemic clearance

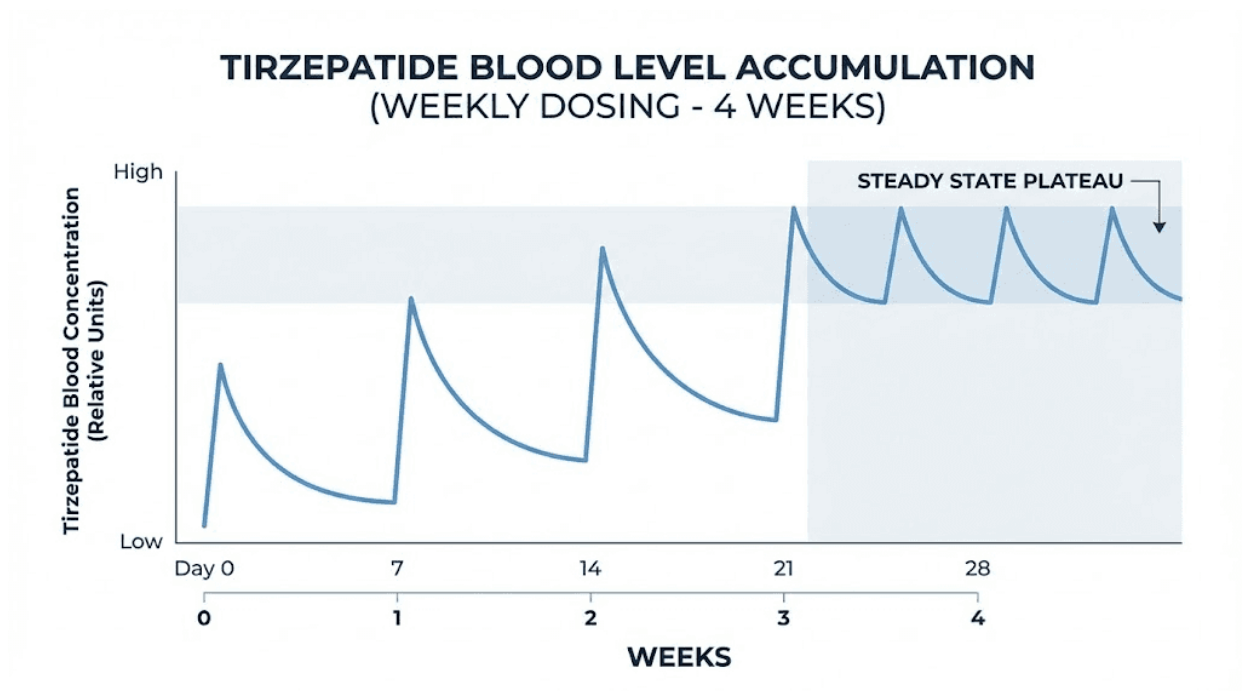


Figure 3: Pharmacokinetic Profile of Tirzepatide Following Once-Weekly Subcutaneous Administration

Source: Adapted from Thomas MK et al. [25]

5. CLINICAL TRIALS AND RESEARCH EVIDENCE

Tirzepatide has been evaluated in two major clinical development programmes: SURPASS (for T2DM) and SURMOUNT (for obesity). These programmes collectively represent the most comprehensive evidence base for a dual incretin agonist to date.^[4,5,6,27,28]

5.1 The SURPASS Clinical Programme (Type 2 Diabetes)

The SURPASS programme comprised five pivotal phase III randomised controlled trials enrolling adults with T2DM inadequately controlled on varying background therapies. All trials used a primary endpoint of change in HbA1c from baseline at 40 weeks (SURPASS-1, 3, 5) or 52 weeks (SURPASS-2, 4).^[4,27]

5.1.1 SURPASS-1

SURPASS-1 was a placebo-controlled, double-blind trial enrolling 478 adults with T2DM on diet and exercise alone. Participants were randomised to tirzepatide 5, 10, or 15 mg weekly versus placebo for 40 weeks.^[27] All three doses significantly reduced HbA1c: -1.87% (5 mg), -1.89% (10 mg), and -2.07% (15 mg) versus $+0.04\%$ for placebo (all $p < 0.0001$). Body weight reductions were -7.0 kg, -7.8 kg, and -9.5 kg, respectively.^[27] The proportion of participants achieving HbA1c $< 7.0\%$ ranged from 87–92% with tirzepatide versus 20% with placebo.

5.1.2 SURPASS-2

SURPASS-2 was an active-controlled, open-label, non-inferiority and superiority trial comparing tirzepatide (5, 10, 15 mg) with semaglutide 1 mg weekly in 1,879 adults with T2DM on metformin background therapy for 40 weeks.^[5] Tirzepatide 5 mg demonstrated non-inferiority, while 10 mg and 15 mg demonstrated superiority over semaglutide 1 mg for HbA1c reduction. HbA1c reductions were -2.01% , -2.24% , and -2.30% for tirzepatide versus -1.86% for semaglutide. Body weight reductions were -7.6 kg, -9.3 kg, and -11.2 kg versus -5.7 kg ($p < 0.001$ for 10 and 15 mg comparisons).^[5]

5.1.3 SURPASS-3

SURPASS-3 enrolled 1,444 adults with T2DM on metformin $\pm$ SGLT-2 inhibitors and compared tirzepatide with insulin degludec (titrated) over 52 weeks.^[28] Tirzepatide 5, 10, and 15 mg reduced HbA1c by -1.93% , -2.20% , and -2.37% versus -1.34% for degludec

($p < 0.0001$ for all). Weight loss with tirzepatide was -7.5 kg to -12.9 kg versus weight gain of $+2.3$ kg with insulin degludec.

5.1.4 SURPASS-4

SURPASS-4 enrolled 2,002 adults with T2DM and elevated cardiovascular risk on oral antidiabetics and compared tirzepatide versus insulin glargine over 52 weeks.^[28] Tirzepatide demonstrated superior HbA1c reduction (-2.24% at 15 mg vs -1.44% for glargine) and weight reductions of up to -11.4 kg. Hypoglycaemia rates were significantly lower with tirzepatide, an important safety advantage over insulin-based therapy.^[28]

5.1.5 SURPASS-5

SURPASS-5 enrolled 475 adults with T2DM already on insulin glargine $\pm$ metformin and added tirzepatide or placebo for 40 weeks.^[27] Tirzepatide significantly reduced HbA1c (-2.11% at 15 mg vs -0.86% for placebo, $p < 0.0001$) and reduced body weight (-6.2 kg vs $+1.9$ kg). Insulin glargine dose was reduced in tirzepatide-treated patients.^[27]

5.2 The SURMOUNT Clinical Programme (Obesity)

The SURMOUNT programme evaluated tirzepatide specifically for chronic weight management in adults with obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$) or overweight ($\text{BMI} \geq 27 \text{ kg/m}^2$) with at least one weight-related comorbidity.^[6,29]

5.2.1 SURMOUNT-1

SURMOUNT-1 was the landmark pivotal trial enrolling 2,539 adults with obesity or overweight without T2DM, randomised to tirzepatide 5, 10, or 15 mg versus placebo for 72 weeks.^[6] The primary endpoints were percentage weight change from baseline and the proportion achieving $\geq 5\%$ weight reduction. Mean weight reductions were -15.0% (5 mg), -19.5% (10 mg), and -20.9% (15 mg) versus -3.1% for placebo.^[6] Approximately 89.8% of participants receiving 15 mg tirzepatide achieved $\geq 5\%$ weight loss. More than one-third (36.2%) achieved $\geq 25\%$ weight reduction with 15 mg, exceeding outcomes observed with any previously approved anti-obesity pharmacotherapy.^[6]

5.2.2 SURMOUNT-2

SURMOUNT-2 enrolled 938 adults with T2DM and obesity, evaluating tirzepatide 10 and 15 mg versus placebo for 72 weeks.^[29] Weight reductions of –13.4% (10 mg) and –15.7% (15 mg) versus –3.3% for placebo ($p<0.001$ for both) were observed. Meaningful improvements in HbA1c, blood pressure, and lipid profiles were also demonstrated.^[29]

5.2.3 SURMOUNT-3

SURMOUNT-3 evaluated the additional benefit of tirzepatide following an intensive lifestyle intervention lead-in phase. Participants who had achieved $\geq 5\%$ weight loss during a 12-week lifestyle programme were randomised to tirzepatide 15 mg or placebo for 72 weeks.^[29] The tirzepatide group achieved an additional –18.4% weight reduction from randomisation versus –2.5% for placebo, demonstrating that tirzepatide provides clinically meaningful weight loss beyond lifestyle interventions.^[29]

5.2.4 SURMOUNT-4

SURMOUNT-4 addressed the long-term maintenance of weight loss with tirzepatide. Participants who had lost $\geq 10\%$ body weight during an initial 36-week tirzepatide treatment period were re-randomised to continue tirzepatide 15 mg or switch to placebo for 52 weeks.^[29] Continued tirzepatide treatment resulted in an additional –5.5% weight reduction, while the placebo group regained +14.8%, highlighting that continued treatment is essential for weight maintenance.^[29]

Table 3: Summary of Major Clinical Trials of Tirzepatide

Trial	N	Duration	Comparator	HbA1c Reduction*	Weight Change*
SURPASS-1	478	40 wks	Placebo	–1.87 to –2.07%	–7.0 to –9.5 kg
SURPASS-2	1,879	40 wks	Semaglutide 1 mg	–2.01 to –2.30%	–7.6 to –11.2 kg
SURPASS-3	1,444	52 wks	Insulin degludec	–1.93 to –2.37%	–7.5 to –12.9 kg
SURPASS-4	2,002	52 wks	Insulin glargine	Up to –2.24%	Up to –11.4 kg
SURPASS-5	475	40 wks	Placebo (add-on)	Up to –2.11%	Up to –6.2 kg
SURMOUNT-1	2,539	72 wks	Placebo	N/A (no T2DM)	–15.0 to –20.9%

SURMOUNT-2	938	72 wks	Placebo (T2DM)	Clinically significant	-13.4 to -15.7%
SURMOUNT-3	806	72 wks	Placebo (post-LSI)	N/A	Additional -18.4%
SURMOUNT-4	670	52 wks	Placebo (maint.)	N/A	+14.8% regain if stopped

*Values represent maximum dose (15 mg unless stated). LSI = lifestyle intervention. HbA1c reductions represent tirzepatide arm vs. comparator. All results statistically significant ($p < 0.001$) unless stated.

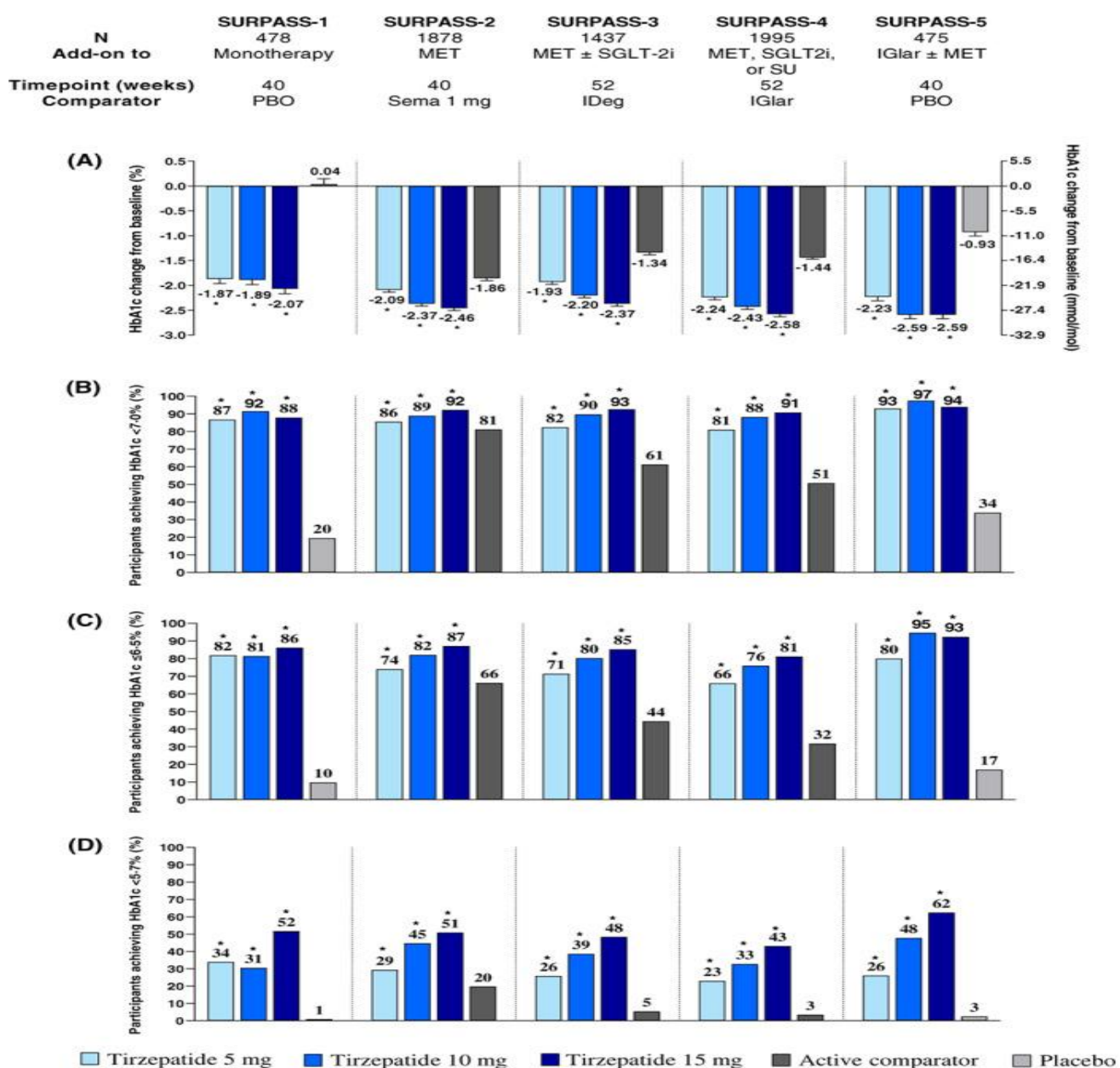


Figure 4: Clinical Development Timeline of Tirzepatide SURPASS Programmes

6. SAFETY PROFILE AND ADVERSE DRUG REACTIONS

The safety and tolerability profile of tirzepatide has been comprehensively characterised across its phase III clinical development programme. The overall safety profile is consistent with the GLP-1 receptor agonist class, with additional considerations attributable to GIP receptor activation.^[31,32]

6.1 Common Adverse Drug Reactions

Gastrointestinal events represent the most frequently reported adverse drug reactions (ADRs) with tirzepatide, particularly during dose escalation. These effects are generally mild-to-moderate in severity and transient in nature.^[31]

- Nausea: reported in 12–18% of patients; most common during initial dose escalation^[31]
- Diarrhoea: reported in 12–14% of patients; typically self-limiting^[31]
- Vomiting: reported in 5–8% of patients^[31]
- Constipation: reported in 6–9% of patients^[31]
- Decreased appetite: common, particularly at initiation^[32]
- Injection site reactions: mild erythema, pruritus in <5% of patients^[32]

6.2 Serious Adverse Drug Reactions

- Pancreatitis: rare but serious; reported in <1% of patients; tirzepatide should be discontinued if pancreatitis is confirmed^[32]
- Gallbladder disease (cholelithiasis, cholecystitis): increased risk associated with rapid weight loss; incidence 0.4–1.1% in trials^[31]
- Hypoglycaemia: risk is low when used as monotherapy; increased risk when co-administered with insulin secretagogues or insulin (11–17% with concurrent insulin)^[28]
- Severe gastrointestinal events: rare cases of gastroparesis reported; caution warranted in those with pre-existing GI motility disorders^[32]
- Heart rate increase: mild elevation of 2–4 bpm observed; clinical significance in high-risk cardiac patients requires monitoring^[31]

6.3 Contraindications

- Personal or family history of medullary thyroid carcinoma (MTC)^[26]
- Multiple Endocrine Neoplasia syndrome type 2 (MEN2)^[26]
- Known hypersensitivity to tirzepatide or any excipient^[26]
- Pregnancy and breastfeeding (insufficient safety data)^[26]

6.4 Drug Interactions

Tirzepatide does not inhibit or induce CYP450 enzymes, minimising pharmacokinetic drug interactions.^[26]

- Gastric emptying delay may affect absorption of orally administered drugs — particularly those with narrow therapeutic windows^[26]
- Clinically significant interaction with warfarin has not been demonstrated but monitoring is advised^[32]
- Co-administration with insulin or sulphonylureas increases hypoglycaemia risk; dose adjustment of these agents is recommended^[28]

Table 4: Summary of Adverse Drug Reactions with Tirzepatide (

ADR Category	Specific ADR	Frequency	Management
GI (Common)	Nausea	12–18%	Dose titration; transient
GI (Common)	Diarrhoea	12–14%	Supportive; self-limiting
GI (Common)	Vomiting	5–8%	Dose titration
GI (Common)	Constipation	6–9%	Dietary fibre; hydration
Endocrine	Hypoglycaemia	Low (<2% monotherapy)	Monitor; adjust co-therapy
Hepatobiliary	Gallbladder disease	0.4–1.1%	Monitor; surgical if indicated
Pancreatic	Pancreatitis	Rare (<1%)	Discontinue immediately
Thyroid	Thyroid C-cell effects	Animal data only	Contraindicated in MTC history
Cardiovascular	Heart rate increase	2–4 bpm	Monitor in high-risk patients

7. THERAPEUTIC APPLICATIONS

The FDA-approved indications for tirzepatide cover both glycaemic management and chronic weight management, with a rapidly expanding evidence base supporting additional cardiometabolic applications.^[12,13]

7.1 Type 2 Diabetes Mellitus

Tirzepatide (Mounjaro®) is approved as an adjunct to diet and exercise to improve glycaemic control in adults with T2DM. Its efficacy across the SURPASS programme establishes it as a highly effective agent for individuals with inadequate glycaemic control on oral antidiabetics, GLP-1 receptor agonists, or insulin regimens.^[4,27,28]

7.2 Obesity and Chronic Weight Management

Tirzepatide (Zepbound®) is approved for chronic weight management in adults with a BMI ≥ 30 kg/m² or ≥ 27 kg/m² with at least one weight-related comorbidity (hypertension, dyslipidaemia, T2DM, obstructive sleep apnoea). The SURMOUNT-1 trial established weight reductions exceeding those of any previously approved pharmacotherapy.^[6,13]

7.3 Metabolic Syndrome and Cardiometabolic Benefits

Tirzepatide produces clinically meaningful improvements in multiple cardiometabolic risk factors including blood pressure reduction (–6 mmHg systolic), triglyceride lowering (–24%), HDL increase, and hepatic fat reduction. These effects make it particularly valuable for patients with metabolic syndrome.^[23,33]

7.4 Emerging Clinical Applications

Ongoing investigations are evaluating tirzepatide in non-alcoholic steatohepatitis (NASH), heart failure with preserved ejection fraction (HFpEF), polycystic ovary syndrome (PCOS), and obstructive sleep apnoea (OSA).^[24,34] The SURMOUNT-OSA trial reported clinically significant reductions in the Apnoea-Hypopnoea Index (AHI) with tirzepatide treatment.^[34]

Table 5: Approved and Emerging Therapeutic Applications of Tirzepatide

Application	Regulatory Status	Key Evidence	Clinical Impact
Type 2 Diabetes	FDA approved (2022)	SURPASS 1–5	HbA1c ↓ up to 2.59%
Obesity/Weight Mgmt	FDA approved (2023)	SURMOUNT 1–4	Weight ↓ up to 22.5%
NASH/MAFLD	Phase II/III (ongoing)	SYNERGY-NASH trial	Hepatic fat ↓, fibrosis

HFpEF	Phase III (SUMMIT)	SUMMIT trial	Functional capacity ↑
Sleep Apnoea	Phase III (SURMOUNT-OSA)	AHI reduction	Symptom improvement
PCOS	Early investigation	Observational studies	Hormonal/metabolic benefit

8. COMPARATIVE ANALYSIS

Tirzepatide's dual incretin mechanism distinguishes it pharmacologically from established GLP-1 receptor agonists. The following comparative analysis evaluates tirzepatide against semaglutide and liraglutide across key pharmacological and clinical parameters.^[5,35,36]

8.1 Pharmacological Comparison

Semaglutide (Ozempic®/Wegovy®) and liraglutide (Victoza®/Saxenda®) act exclusively on GLP-1 receptors. Tirzepatide's additional GIP receptor agonism confers superior insulin secretion, enhanced adipose tissue effects, and potentially improved GI tolerability.^[17,35]

Table 6: Comparative Pharmacological and Clinical Profile: Tirzepatide vs. Semaglutide vs. Liraglutide

Parameter	Tirzepatide	Semaglutide	Liraglutide
Mechanism	Dual GIP + GLP-1 agonist	GLP-1 agonist	GLP-1 agonist
Half-Life	~5 days	~7 days	~13 hours
Dosing	Once weekly (SC)	Once weekly (SC) / daily (oral)	Once daily (SC)
Max Reduction HbA1c	~2.59%	~1.86% (1 mg SC)	~1.6%
Max Reduction Weight	~22.5% (SURMOUNT)	~14.9% (STEP 1)	~8% (SCALE)
Cardiovascular Outcome	SURPASS-CVOT ongoing	Proven benefit (SUSTAIN-6)	Proven benefit (LEADER)
GI Tolerability	Moderate; GIP may attenuate	Moderate	Higher incidence
Hypoglycaemia Risk	Low (monotherapy)	Low (monotherapy)	Low (monotherapy)
FDA Approval	T2DM + Obesity	T2DM + Obesity	T2DM + Obesity
Cost	Higher	High	Moderate-High

The SURPASS-2 trial provides the most direct head-to-head comparison, demonstrating superior HbA1c and weight reductions with tirzepatide 10 and 15 mg compared to semaglutide 1 mg.^[5] A dedicated cardiovascular outcomes trial (SURPASS-CVOT) is

ongoing to establish cardiovascular risk reduction data equivalent to that available for semaglutide and liraglutide.^[37]

9. ADVANTAGES AND LIMITATIONS

Table 7: Summary of Advantages and Limitations of Tirzepatide

Advantages	Limitations
Novel dual GIP/GLP-1 receptor agonism — first-in-class mechanism [1,2]	High acquisition cost — limits accessibility in LMICs [38]
Superior HbA1c reduction versus all approved comparators [4,5]	Gastrointestinal side effects — nausea, vomiting, diarrhoea [31]
Unprecedented weight loss up to 22.5% in obesity trials [6]	Long-term cardiovascular outcome data not yet available [37]
Once-weekly subcutaneous dosing — improves adherence [26]	Black box warning for thyroid C-cell tumours [26]
Low hypoglycaemia risk in monotherapy [31]	Contraindicated in pregnancy and MEN2 [26]
Multi-system metabolic benefits (BP, lipids, liver fat) [23]	Drug supply shortages reported globally [38]
Potential emerging indications (NASH, HFpEF, OSA) [24,34]	Limited real-world long-term safety data >2 years [32]

10. FUTURE PERSPECTIVES

Tirzepatide represents the vanguard of a new therapeutic paradigm in metabolic medicine. Multiple ongoing investigations and emerging research directions are poised to further expand its clinical utility.^[37,39]

10.1 Cardiovascular Outcome Trial

The SURPASS-CVOT trial (NCT04255433) is a dedicated cardiovascular outcomes trial evaluating whether tirzepatide reduces major adverse cardiovascular events (MACE — cardiovascular death, non-fatal myocardial infarction, non-fatal stroke) in adults with T2DM and established cardiovascular disease.^[37] Results are anticipated by 2026–2027 and are expected to be pivotal for positioning tirzepatide alongside semaglutide and liraglutide in guidelines for high-risk cardiovascular patients.^[37]

10.2 Expanded Indications

The SYNERGY-NASH trial (Phase II) demonstrated significant reductions in liver fat fraction and hepatic inflammation with tirzepatide in NASH patients, with phase III trials underway.^[24] The SURMOUNT-OSA trial demonstrated clinically meaningful reductions in AHI scores, supporting FDA review for sleep apnoea indication.^[34] Heart failure with preserved ejection fraction (SUMMIT trial) has demonstrated improvements in functional capacity with tirzepatide.^[40]

10.3 Triple Agonists and Next-Generation Agents

Building on tirzepatide's success, next-generation triple agonists targeting GIP, GLP-1, and glucagon receptors (e.g., retatrutide, Eli Lilly) have entered clinical trials. Phase II data for retatrutide demonstrated weight reductions of up to 24.2% at 48 weeks — exceeding tirzepatide — with additional thermogenic benefits from glucagon receptor activation.^[39]

10.4 Personalised Medicine and Biomarkers

Future research is expected to identify predictive biomarkers for tirzepatide response, including GIP receptor polymorphisms, adipokine profiles, and gut microbiome signatures.^[41] This precision medicine approach could enable optimal patient selection and individualised dose titration strategies.^[41]

10.5 Oral Formulations

Ongoing research into oral formulations of GIP/GLP-1 agonists, analogous to oral semaglutide (Rybelsus®), may eventually extend tirzepatide's accessibility and patient preference beyond injectable administration.^[42]

10.6 Research Gaps

- Long-term cardiovascular outcome data (>5 years) are not yet available^[37]
- Head-to-head data against semaglutide 2.4 mg (Wegovy) for obesity endpoints are lacking^[35]
- Safety and efficacy in paediatric populations require dedicated investigation^[32]
- Cost-effectiveness analyses in global healthcare systems are limited^[38]
- Potential drug holiday strategies and optimal treatment duration remain undefined^[29]

11. CONCLUSION

Tirzepatide constitutes a landmark advancement in pharmacological management of type 2 diabetes mellitus and obesity. As the first-in-class dual glucose-dependent insulintropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptor agonist, it exploits synergistic incretin biology to achieve therapeutic outcomes that transcend those of any single-receptor agonist currently available.^[1,2]

The SURPASS clinical programme demonstrated consistent, statistically superior reductions in HbA1c (up to 2.59%) and body weight (up to 11.4 kg) compared to placebo and active comparators including semaglutide 1 mg, insulin degludec, and insulin glargine, across diverse T2DM patient populations.^[4,5,27,28] The SURMOUNT programme established tirzepatide as the most effective pharmacological weight management agent currently approved, achieving up to 22.5% body weight reduction — outcomes previously attainable only through bariatric surgery in select populations.^[6,29]

Its pharmacokinetic profile — characterised by approximately 80% subcutaneous bioavailability, >99% albumin binding, a ~5-day half-life, and absence of CYP450 metabolism — supports convenient once-weekly dosing with minimal drug interaction potential.^[25,26]

The safety profile, while consistent with the GLP-1 receptor agonist class, is generally acceptable. Gastrointestinal adverse effects predominate but are predominantly mild-to-moderate and transient, particularly with the recommended stepwise dose escalation protocol.^[31,32] The absence of significant hypoglycaemia in monotherapy represents an important advantage over older antidiabetic agents.^[31]

Challenges remain, including high treatment costs, global supply limitations, absence of long-term cardiovascular outcome data, and lack of head-to-head data against higher-dose semaglutide for obesity management.^[37,38] Ongoing trials in NASH, HFpEF, obstructive sleep apnoea, and renal disease promise to substantially broaden tirzepatide's therapeutic applicability.^[24,34,40]

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